

*I want to thank my family for the continued support throughout university,
and any friend I met during my time at Bocconi.*

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Chapter 1

Introduction

1.1 Motivation

Financial time series data is never fully observed. Limit-order book feeds drop quotes during bursts of order cancellations. Equity markets suspend trading when prices move too violently. Over-the-counter bond markets leave entire yield-curve tenors unquoted for days. Implied volatility surfaces contain large regions where no liquid option trade has occurred. Macro and alternative data arrive at mixed, irregular frequencies, with publication lags that create structural gaps at the daily level. This missing data is pervasive and often elusive, and gets in the way of advanced analysis. Thus practitioners and researchers face the same decision: what value should be attributed to the missing observation, and how much uncertainty should be attached to that attribution?

The simplest and most common solution is to carry forward the last observed value, but this can lead to systematic mispricing, underestimation of risk, and distortion of correlations among assets.

We aim to take a different path, one that looks for solutions preserving as much as possible the statistical properties of the series, in order to maintain the integrity of the underlying data and consequently the robustness of any downstream model that relies on it. Since this can be considered an inference problem, we also explore machine learning algorithms, and, as is usual in that literature, we focus on the quality of the inference rather than on the interpretability of the model.

In this thesis we test these different methods of imputation and evaluate them not only on the accuracy of the imputation itself, through metrics such as MSE, but also on the quality of the downstream model that uses the imputed data, and on how far that model deviates from the one obtained when all the data is observed.

1.2 Literature review

The literature on missing data is vast and spread across statistics, econometrics and, more recently, machine learning. Rather than attempting an exhaustive survey, we trace the ideas this thesis directly builds on: the foundational framework that defines what missingness is, the classical likelihood-based methods with the EM algorithm at their core, the progressive shift towards algorithmic and deep learning imputers, and finally how these tools have been absorbed by economics and finance.

1.2.1 Foundations

Dropping incomplete records, or filling them with some convenient value and proceeding as if nothing had happened, has always been common practice. It was not, however, the only option available, and the problem was attacked seriously long before it acquired its modern formulation. In the design-of-experiments tradition of the 1930s an incomplete field trial was already handled by an explicit estimate of what had not been observed. Allan and Wishart (1930) gave the first missing plot formula: for a randomised block design they derived the value which, once inserted in the empty cell, leaves the treatment comparisons of the analysis of variance unbiased. Yates (1933) extended the device to any number of missing plots and to more general layouts, and identified the principle behind it, namely that the substituted values are exactly those that minimise the residual sum of squares. He also observed that the degrees of freedom must be reduced by the number of values supplied, since a filled cell carries no information of its own; the same warning applies to every single imputation in this thesis. Bartlett (1937) then recast the computation as an analysis of covariance with one dummy covariate per missing observation, which returns the estimate and the corrected residual sum of squares in a single pass.

Outside designed experiments the work started just as early. Wilks (1932) studied the moments of estimates computed from fragmentary samples of a multivariate normal population, Anderson (1957) later obtained closed-form maximum likelihood estimates when the pattern of missingness is nested, and Buck (1960) proposed filling each gap with a regression prediction from the variables observed on the same unit, a method he wrote for use on an electronic computer. Computers also turned the iterative version of Yates's method into a general algorithm: Healy and Westmacott (1956) start from a guess for the missing entries, fit the model by least squares, replace the entries by their fitted values, and iterate until the values stabilise. Dempster et al. (1977) would later list this scheme among the special cases their algorithm unifies. By the time Afifi and Elashoff (1966) reviewed the field there was already a substantial body of techniques, and Orchard and Woodbury (1972) had extracted from it a missing information principle of some generality. What this work lacked was a language for the missingness itself. Every procedure carried an implicit assumption about why the entries were absent, the assumption was rarely

stated, and there was no way to say when a procedure was justified and when it was not.

The turning point is Rubin (1976), who supplied that language by treating missingness itself as a random object. Given a data matrix and an indicator matrix flagging which entries are observed, the joint distribution factorises into the law of the data and the law of the *missingness mechanism*, the conditional distribution of the indicators given the data. Rubin’s classification of this mechanism is still the standard language of the field: data are missing completely at random (MCAR) when the mechanism is independent of the data; missing at random (MAR) when it depends only on the observed part; and missing not at random (MNAR) otherwise. The central result of the paper is *ignorability*: under MAR, plus a parameter-distinctness condition, likelihood-based and Bayesian inference can ignore the mechanism altogether and work with the observed-data likelihood. This single result justifies most of the methods in this thesis; whenever it fails, as in the stressed missingness pattern of Section 2.1.1, every method that relies on it stands on shaky ground.

The monograph of Little and Rubin (2019), first published in 1987, organised the field around this framework and remains its canonical reference. Their taxonomy of approaches (complete-case analysis, weighting, single imputation, likelihood-based methods) is still the way the subject is taught, and their analysis of the simple methods is directly relevant here: listwise deletion is unbiased only under MCAR and can discard a dramatic fraction of the sample, while unconditional mean imputation distorts variances and correlations even under MCAR. These are precisely the two baselines we adopt in Chapter 2, and the literature predicts their failure modes quite exactly.

Two other foundational strands deserve mention. The first is weighting: instead of imputing, one reweights the complete cases by the inverse of their probability of being observed. Robins et al. (1994) developed this idea into a full semiparametric theory, including doubly robust estimators; Seaman and White (2013) give an accessible review. The second is *multiple imputation*, introduced by Rubin (1987): draw several imputed datasets from the posterior predictive distribution of the missing values, analyse each one as if it were complete, and combine the results with simple rules that propagate the imputation uncertainty into the final standard errors (see also Rubin, 1996). Multiple imputation is the historical answer to a criticism that applies to every single-imputation method in this thesis: an imputed dataset treated as real understates uncertainty, because imputed values are estimates, not observations.

The MNAR case is fundamentally harder, since the mechanism is not identifiable from the observed data alone and any analysis rests on untestable assumptions. In econometrics the problem was attacked early and largely independently by Heckman (1979), whose sample-selection model corrects for non-randomly missing outcomes through an explicit model of the selection process; it remains the standard MNAR tool in applied economics. In biostatistics the accepted practice is sensitivity analysis across plausible mechanisms (Molenberghs and Kenward, 2007). Finally, the translation of MAR to

dependent, dynamically evolving data is more subtle than it appears; Commenges and Jacqmin-Gadda (2022) give a rigorous stochastic-process formulation, which is the natural setting for financial time series.

1.2.2 Likelihood methods and the EM algorithm

Once ignorability is granted, inference reduces to maximising the observed-data likelihood, which almost never has a closed form because the missing entries must be integrated out. The Expectation–Maximisation algorithm of Dempster et al. (1977) is the fundamental tool for this task, and arguably the most important paper in the field after Rubin (1976). Its logic mirrors the structure of the problem: the E-step computes the conditional expectation of the complete-data log-likelihood given the observed data and the current parameters, and the M-step maximises it. The paper proved that each iteration increases the observed likelihood, unified a large number of pre-existing special-purpose schemes, and named the method. The convergence theory was completed by Wu (1983), who fixed a flaw in the original proof and gave the regularity conditions under which EM converges to a stationary point. Since EM never forms the observed information matrix, standard errors require extra work; Louis (1982) showed how to recover them from complete-data quantities. Among the many refinements that followed, the ECM and ECME algorithms (Liu and Rubin, 1994; Meng and Rubin, 1993) replace the M-step with cheaper conditional maximisations; McLachlan and Krishnan (2008) is the standard book-length treatment.

For our purposes the relevant instance is EM for the mean vector and covariance matrix of an incomplete multivariate normal sample (Little and Rubin, 2019, Ch. 11): the E-step reduces to linear regressions of the missing coordinates on the observed ones, and the fixed point delivers both the maximum likelihood estimate of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and, as a by-product, conditional-expectation imputations. Since daily returns are decidedly non-Gaussian (Cont, 2001), the robust variant matters: Little (1988) derived EM for the multivariate Student- t , whose E-step simply downweights outlying observations, and Liu and Rubin (1995) made its estimation practical via ECME. The implementation used in Chapter 2 is the Gaussian one; the robust variant is discussed there as the natural refinement, and Chapter 4 returns to it.

The time series counterpart of this machinery is the state-space framework. The Kalman filter (Kalman, 1960) accommodates missing observations with no modification at all: when a measurement is absent, the update step is skipped and the prediction carries through. Jones (1980) exploited this to fit ARMA models to incomplete series by exact maximum likelihood, and Shumway and Stoffer (1982) closed the loop with EM: the Kalman smoother computes exactly the conditional expectations the E-step requires, so parameter estimation and imputation come out of the same iteration. Harvey and Pierse (1984) applied these ideas to economic time series, Kohn and Ansley (1986) settled exact

interpolation for ARIMA models, and the monographs of Harvey (1989) and Durbin and Koopman (2012) remain the references for the general theory.

1.2.3 From multiple imputation to machine learning

On the applied side, the 1990s and 2000s standardised imputation practice. Schafer (1997) provided workable algorithms for multiple imputation under joint multivariate models, and the chained equations approach of van Buuren and Groothuis-Oudshoorn (2011) replaced the joint model with a sequence of univariate conditional models, one per variable, cycled until convergence; its flexibility made it the de facto standard in biostatistics and the social sciences. van Buuren (2018) gives a modern account of the whole practice.

In parallel, a more algorithmic tradition developed methods that dispense with an explicit probability model. k -nearest-neighbours imputation was popularised by Troyanskaya et al. (2001) for gene expression data and transfers directly to return panels: a missing entry is filled with a weighted average of the corresponding entries of similar rows (dates, in our case); random forests followed (Stekhoven and Bühlmann, 2012). A distinct and elegant line is low-rank matrix completion (Candès and Recht, 2009; Mazumder et al., 2010), which recovers missing entries under the assumption that the data matrix is approximately low-rank, a natural assumption for equity returns where a few factors explain most of the covariance.

Deep learning entered the field once recurrent networks were adapted to irregular inputs. GRU-D (Che et al., 2016) feeds the missingness mask and the time since the last observation into a gated recurrent unit with learned decay; BRITS (Cao et al., 2018) treats imputations as variables of a bidirectional recurrent graph, trained by consistency between the two directions, and requires no assumption on the generating process. Generative approaches recast imputation as conditional sampling: GAIN adapts adversarial training (Yoon et al., 2018), GP-VAE combines a variational autoencoder with a Gaussian process prior in latent space (Fortuin et al., 2020), SAITS relies on self-attention (Du et al., 2023), and CSDI (Tashiro et al., 2021) uses conditional score-based diffusion, producing a full predictive distribution for the missing values rather than a point estimate, which is why we select it as our generative representative. Dong et al. (2024) survey this rapidly growing area, and Hippert-Ferrer et al. (2025) bridge it with the signal processing literature.

Two caveats recur across this literature. First, essentially all deep imputers assume MAR, usually implicitly through their masked training objective; identification under more general mechanisms has been studied only very recently (Ma et al., 2025). Second, evaluation is almost universally pointwise, RMSE or MAE on artificially masked entries, which says little about whether the joint distribution, and in particular the covariance structure, survives imputation. Both caveats are starting points for this thesis.

1.2.4 Applications in economics and finance

In economics, missing data rarely goes by that name. The oldest example is non-synchronous trading: Scholes and Williams (1977) showed that stale prices, which are in effect missing prices carried forward, bias beta estimates, and derived corrected estimators. This is exactly the last-observation-carried-forward distortion described in Section 1.1.

The macroeconometric literature turned missing data handling into routine practice, because macro datasets are incomplete by construction: series are released at mixed frequencies and with staggered publication lags (the so-called ragged edge). Stock and Watson (2002) used an EM–principal-components iteration to extract factors from large unbalanced panels; Mariano and Murasawa (2003) treated quarterly GDP as a monthly series with two out of three values missing, estimated in state-space form; Giannone et al. (2008) formalised nowcasting, where the Kalman filter absorbs data releases as they arrive; and Doz et al. (2012) and Bańbura and Modugno (2014) developed maximum likelihood estimation of large factor models under arbitrary missingness patterns, essentially scaling Shumway and Stoffer (1982) to hundreds of series. This machinery runs in production at central banks today. For panels with a time dimension, Honaker and King (2010) adapted multiple imputation to time-series cross-section data, noting that off-the-shelf procedures ignore temporal smoothness.

In finance proper, the seminal reference is Stambaugh (1997), who considered assets whose return histories start at different dates, the typical shape of any real multi-asset panel, and derived maximum likelihood estimators of means and covariances that use the long histories to sharpen the moments of the short ones. The message, that discarding incomplete history throws away information about the covariance matrix, is a direct ancestor of our evaluation criteria. Kofman and Sharpe (2003) later imported multiple imputation into empirical finance, showing on several canonical applications that complete-case analysis materially biases the conclusions.

The topic has recently returned to the centre of asset pricing, driven by the firm-characteristics panels used in cross-sectional factor models, where the majority of entries can be missing. Bryzgalova et al. (2025) document that the profession’s ad hoc fixes, typically cross-sectional medians, are not innocuous: the choice of imputation changes the estimated risk premia, and they propose a factor-based imputation exploiting both cross-sectional and time series dependence. In parallel, Freyberger et al. (2025) show that firm characteristics are missing *not* at random and develop estimators that remain consistent under such mechanisms. Both papers make, in the cross-section of characteristics, the point this thesis makes for return panels: what matters is not the imputation error itself but its effect on the downstream object of interest. Finally, Choudhury et al. (2021) highlight a specifically financial pitfall: imputing with a model trained on the full sample leaks future information into the imputed values, and they formalise the resulting trade-off

between look-ahead bias and imputation variance.

Summing up, the statistical literature provides the taxonomy of mechanisms and the classical estimators, the machine learning literature provides increasingly powerful conditional models trained under an implicit MAR assumption, and the finance literature provides both the warning that missingness is often MNAR precisely when it matters most (crises, illiquidity) and the reminder that imputation should be judged by the downstream model. The experimental design of this thesis, controlled injection of MCAR and MNAR patterns into a complete equity panel, classical and deep imputers side by side, and evaluation on covariance geometry, risk measures and portfolio construction rather than pointwise error alone, sits deliberately at the intersection of these three strands.

1.3 Research questions

The thesis is organised around four questions.

1. **What drives the damage, the amount of missing data or the mechanism that produced it?** We compare patterns that remove entries at random with a pattern that removes them preferentially in turbulent periods, at comparable shares of missing cells.
2. **Do the deep imputers earn their cost?** Recurrent and diffusion-based models are the current state of the art on imputation benchmarks. We ask whether they improve on classical likelihood and low-rank estimators on a financial panel, and what any improvement is worth once training and sampling time are counted.
3. **Is pointwise accuracy a good proxy for the quality of the imputed panel?** Reconstruction error is the standard criterion, but no downstream model consumes individual cells. We ask whether a method that fills the holes accurately also preserves the covariance matrix, the tails, the marginal distributions and the temporal dependence of the returns.
4. **Do the rankings survive downstream?** We ask whether the ordering of the methods given by the reconstruction metrics is the same ordering given by the decisions a user would take on the imputed panel, and where the two disagree.

1.4 Contributions

1. A controlled testbed for imputation on financial data: a complete panel of daily log returns for fifty S&P 500 constituents, three injection mechanisms at five severities each, and seven imputers from deletion to a conditional diffusion model, all implemented and run within a single framework.

2. An evaluation suite that goes past reconstruction error. Alongside MAE and RMSE we score the covariance matrix, the five percent Value-at-Risk, the marginal distributions in Wasserstein distance, and two temporal statistics, the autocorrelation of absolute returns and the leverage curve, which the permutation-invariant metrics cannot detect.
3. The downstream evaluations, in which an imputation is judged by the object a user would build on it, referenced against the same object built on the uncorrupted panel. We instantiate the idea twice, with the tangency portfolio and with the factor structure of the panel, and design both to rank the reconstruction methods against one another rather than against a naive baseline.
4. The empirical findings themselves, among them that pointwise and structural quality can diverge by close to an order of magnitude, that in this experiment the neural models do not improve on EM at a computational cost several orders of magnitude higher, that under the tested MCAR configurations every conditional imputer inflates the factor structure of the panel, as point imputations built from conditional location estimates are expected to, and that the ranking of the methods is least stable exactly where the data is missing not at random.

1.5 Outline of the thesis

Chapter 2 describes the data, the three injection mechanisms, the seven imputation models and the evaluation criteria, both the pointwise and structural metrics and the two downstream evaluations. Chapter 3 reports the results, one table per injection pattern, then the portfolio and the factor-structure evaluations, and closes with the computational cost of each method. Chapter 4 returns to the four questions above and states what remains open.

Chapter 2

Methodology

As already mentioned, missing data can arise in countless different ways, and the choice of methodology depends on the nature of the missingness, the type of data, and the specific goals of the analysis. We will focus on a subset of cases, namely those that involve imputation of the missing values with information drawn from the whole dataset, with the goal in mind of training a downstream model that is as close as possible to the one that would have been trained on the complete dataset. The setting is therefore an offline reconstruction one: no causal constraint is imposed, and every method is free to use the full sample, including observations that fall after the date being filled. Our choice of data, models, and evaluation methods will be guided by these constraints. We will nevertheless mention when/how models should be tweaked to solve adjacent problems.

2.1 Data

As our main baseline, we chose the most studied and reliable dataset in finance: the end-of-day prices of the S&P 500 constituents, fetched with `yfinance` from 1/1/2010 to 30/5/2026. This is not a series that suffers badly from missing data; the only structural gaps are occasional circuit breaks and listings or delistings from the index. To obtain a full dataset with no "real" missing data, we removed each stock that contained at least one missing observation. Furthermore, we selected a sample of 50 stocks from the index, stratified across industries; this is a sensible choice because:

1. with more than 400 stocks the covariance matrix Σ quickly becomes unreliable (noisy covariances, skewed eigenvalues, unstable inversion) as the ratio of observations to assets falls;
2. it is easier to verify that the reference data is actually correct;
3. model training requires considerably less memory and computing time;

4. a cross section of fifty assets is already wide enough to behave like a realistic multi-asset problem, with a market factor, industry structure and a substantial amount of diversifiable idiosyncratic risk, rather than a handful of series.

This dataset will serve as the ground truth against which we will compare our results. In parallel, we will modify it by inserting fake missing observations according to different rules, as we define in the subsequent sections.

An important choice is whether to operate in the price space P_t or in the log return space R_t . There are advantages and disadvantages to each. The raw dataset is prices, and in practice missing data mostly arrives as missing prices; in most applications we also care slightly more about prices than returns, since the former are what is actually bought and sold. But P_t is constrained to the positive half-line ($p_{t,k} > 0$) and is strongly level dependent, which makes the series non-stationary and difficult for machine learning algorithms to handle. We therefore work in R_t .

This choice carries an abstraction that is worth stating plainly, because the motivation of Section 1.1 is phrased in terms of prices while the experiments below are not. The objects we delete and reconstruct are entries of the return panel, not prices. Deleting a return is a stylised representation of a missing price rather than a literal one: carrying a price forward across a gap corresponds to a zero return over that gap, which the zero imputation of Section 2.2.2 reproduces exactly, but a real stale price is also followed by a catch-up return once quoting resumes, which absorbs the move that was never recorded. Working in return space suppresses that catch-up effect, so the results below should be read as statements about the return panel a downstream model consumes, not as a full model of how stale prices propagate.

2.1.1 Missingness injection

We used three algorithms to simulate different missingness patterns:

1. **Sparse**: each scalar $r_{t,k}$ is missing with probability r , independent of the rest of the dataset. This is the easiest example of MCAR and can simulate a randomly corrupted dataset;
2. **Contiguous gaps**: n seed cells $r_{t,k}$ are drawn uniformly at random, and each seed is extended forward into a run $r_{t,k}, r_{t+1,k}, \dots, r_{t+L-1,k}$ of the same asset, whose length L is drawn from an exponential distribution with mean ℓ and rounded down to a whole number of trading days. The second parameter of a gap level is therefore the mean run length ℓ , measured in trading days, and not a decay rate; a level is written (n, ℓ) throughout. Runs are drawn independently and may overlap, so the realised share of missing cells is slightly below $n\ell/(TN)$. This is still an example of MCAR and simulates assets failing to trade because of random shortages of liquidity;

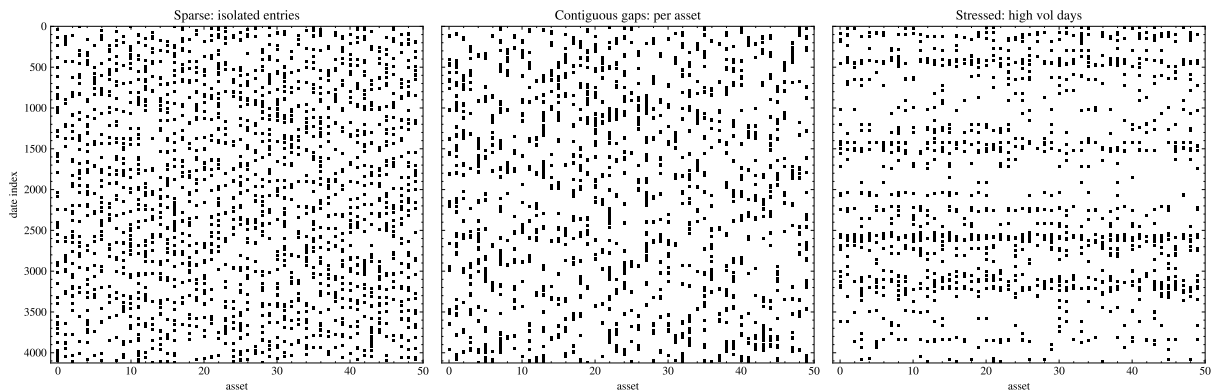


Figure 2.1: Examples of the three missingness patterns on the sampled panel. Each marker is a deleted entry, with dates on the vertical axis and assets on the horizontal one. Sparse missingness (left) removes isolated cells, contiguous gaps (centre) remove runs of consecutive dates within a single asset, and the stressed pattern (right) removes entries preferentially on the most volatile dates, which aligns the deletions into horizontal bands across the cross section.

3. **Stressed:** for each date t we form the equally weighted average log return of the fifty sampled assets, $\bar{r}_t = N^{-1} \sum_k r_{t,k}$, which serves as our market proxy, and score the date by its square $\bar{r}_t^2$. This is what we mean by the index squared return: no external index series is used, the proxy is built from the panel itself. Dates whose score falls at or above the p quantile of that score are declared stressed, and on those dates every entry is deleted independently with probability h , while on the remaining dates entries are deleted with probability r . A level is written (r, h, p) . This simulates the shortages of liquidity and the circuit breaks that occur when markets are volatile.

The scoring statistic is computed on the complete panel, and the values it depends on are then among those hidden from the imputer. The market proxy is not retained as a separately observed series either: on a stressed date the imputer sees only the entries that survived, from which $\bar{r}_t$ cannot be recovered. Relative to the information set actually available to the imputation methods, the mechanism is therefore non-ignorable, and we treat it throughout as a stylised MNAR pattern. We note that the same construction would be MAR conditional on the score if that score were handed to the imputer as a fully observed covariate; it is the fact that it is not, rather than anything about the functional form, that makes the pattern non-ignorable here.

For each of the three patterns we vary the parameters above to obtain five levels of missingness. The levels are chosen so that the expected share of missing cells is the same at the corresponding level of each pattern, which is what allows the three sets of results to be compared row block by row block: any difference between them is attributable to the mechanism rather than to the amount of data removed. The exact values are reported in Table 2.1, whose last column gives the expected missingness ratio common to the three

patterns at that level.

Table 2.1: Parameters of the three injection mechanisms at the five levels. Sparse levels are the per-cell deletion probability r ; gap levels are the number of seeds n and the mean run length ℓ in trading days; stressed levels are the background rate r , the deletion probability h on stressed dates and the quantile p of the score above which a date counts as stressed. The last column is the expected share of missing cells, which the levels are matched on.

Sparse (r)	Gaps (n, ℓ)	Stressed (r, h, p)	Expected rate
0.001	(20, 10)	(0, 0.1, 0.99)	0.001
0.01	(200, 10)	(0.005, 0.5, 0.99)	0.01
0.05	(1000, 10)	(0.04, 0.2, 0.95)	0.05
0.1	(1000, 20)	(0.05, 0.5, 0.9)	0.1
0.3	(2000, 30)	(0.2, 1, 0.9)	0.3

2.2 Models

2.2.1 Deletion

The baseline of complete-case analysis: every date containing at least one missing value is dropped, and all downstream estimation runs on the surviving complete rows. Under MCAR the retained dates are a random subsample of the panel, so moment estimates remain consistent (Little and Rubin, 2019); the cost is statistical efficiency, and in the cross-section it is severe. With sparse missingness at rate r , a date survives with probability $(1 - r)^N$: at $r = 0.05$ and $N = 50$ assets, fewer than 8% of the dates survive. Under the stressed pattern the damage changes nature rather than size: the deleted dates are exactly the turbulent ones, so the surviving subsample is calm by construction and every risk measure computed from it is biased downwards, no matter how much data is left.

2.2.2 Zero imputation

The simplest imputation that preserves the shape of the panel sets every missing return to zero. Since daily log returns have a mean of essentially zero, this is also the unconditional location estimate to within a negligible amount, and it is exactly the reconstruction implied by carrying the last observed price forward across the gap. The tables of Chapter 3 report it under the name Zero.

Its failure mode is predictable in closed form. Every second moment is damped: under sparse missingness at rate r , a variance term survives whenever its own entry is observed while a covariance term needs both entries at once, so estimated volatilities scale by $\sqrt{1 - r}$ and pairwise correlations by $(1 - r)$, shrinking the covariance matrix towards its diagonal. Under the stressed pattern the distortion is stronger still, because the entries

being replaced are on average the largest ones. This is precisely the failure the structural metrics of Section 2.3 are designed to expose.

2.2.3 Low-rank matrix completion

A few pervasive factors describe most of the covariance of equity returns, so the return panel is close to a low-rank matrix plus idiosyncratic noise. Matrix completion turns this observation into an imputation device: if the underlying matrix has rank $q \ll N$, the missing entries can be recovered from the observed ones – exactly so, under conditions on the missingness pattern and on how spread out the factors are (Candès and Recht, 2009). We adopt the iterative-SVD scheme of Mazumder et al. (2010): standardise each asset on its observed entries, fill the missing cells with zeros, then alternate between projecting the completed panel onto the set of rank- q matrices (truncated SVD) and restoring the observed entries, until the panel stops changing. The fixed point fills every missing cell with a q -factor reconstruction estimated on the fly, without ever specifying the factors. The nuclear-norm variant of the same algorithm, which soft-thresholds the singular values instead of truncating them, shrinks every imputed entry towards zero and thereby inherits a milder version of the covariance damping of zero imputation; we therefore keep the hard rank constraint, with q small, in line with the number of pervasive equity factors.

2.2.4 EM

Our likelihood-based model treats the dates as i.i.d. draws $r_t \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and maximises the resulting observed-data likelihood with the EM algorithm of Dempster et al. (1977). Write $r_t = (r_{t,o}, r_{t,m})$ for the observed and missing parts of date t (the split changes from date to date). The E-step fills each missing block with its conditional expectation,

$$\mathbb{E}[r_{t,m} \mid r_{t,o}] = \boldsymbol{\mu}_m + \boldsymbol{\Sigma}_{mo} \boldsymbol{\Sigma}_{oo}^{-1} (r_{t,o} - \boldsymbol{\mu}_o),$$

while the M-step re-estimates $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ from the completed panel, adding back the conditional covariance $\boldsymbol{\Sigma}_{mm} - \boldsymbol{\Sigma}_{mo} \boldsymbol{\Sigma}_{oo}^{-1} \boldsymbol{\Sigma}_{om}$ of the cells just filled – the correction that separates EM from naively iterating a regression fill, which would understate the covariance. At convergence we obtain the maximum likelihood estimate $(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}})$ and, as a by-product, imputations that are conditional expectations, and therefore the fills that minimise the expected squared error given the observed part of the date. In the implementation, dates sharing the same missingness pattern are grouped together, so each pattern costs a single linear solve per iteration. A small ridge term is added to the covariance and to each observed block before it is inverted, which keeps the solves stable without materially shrinking the estimate. Daily returns are of course heavier-tailed than the Gaussian assumption wants; the same iteration extends to a multivariate Student- t , whose E-step

gains a per-date weight, decreasing in the Mahalanobis distance of the observed block, that discounts extreme dates in the update of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ (Little, 1988; Liu and Rubin, 1995). That robust variant is not used here: all EM results reported in Chapter 3 come from the Gaussian model described above, and the Student- t extension is noted only as the natural next refinement.

2.2.5 k -nearest-neighbour imputation

k -nearest-neighbour imputation, first popularised on gene-expression data by Troyanskaya et al. (2001), is the donor-based, fully non-parametric member of our set; the tables of Chapter 3 report it as KNN. To fill date t , its distance to every other date is computed as the Euclidean distance over the assets both dates observe, rescaled by the number of co-observed assets so that sparser rows are not mechanically favoured; the missing return $r_{t,i}$ is then the inverse-distance-weighted average of $r_{s,i}$ over the k closest dates s that do observe asset i . Nothing about the scheme is temporal: donors are selected by cross-sectional similarity, wherever they sit in the sample. No distributional assumption is needed, but the method degrades exactly where donors become unreliable: under heavy missingness the distances rest on few co-observed assets, and under the stressed pattern the extreme dates that most need filling have no close, well-observed counterparts to borrow from.

2.2.6 BRITS

BRITS (Cao et al., 2018) is our recurrent neural representative. A window of the panel is processed by two LSTMs, one per time direction. At each step the hidden state is first decayed by a factor learned from the time elapsed since each asset was last observed, so that the memory of a stale reading fades, and two candidate fills are produced: a history-based estimate regressed from the hidden state, and a cross-sectional estimate regressed from the other assets at the same date. A learned gate blends the two, the blend replaces the missing entries, and the completed vector is fed to the next step, so the imputations themselves carry gradients through the network. Training is self-supervised: the network learns to reconstruct the entries that are observed, with a penalty tying the forward and backward passes together, and the missing cells are then read out of the trained network. No explicit model of the returns or of the missingness mechanism is assumed.

2.2.7 CSDI

CSDI (Tashiro et al., 2021) adapts denoising diffusion models (Ho et al., 2020) to imputation. In a diffusion model, data is progressively corrupted by Gaussian noise over many small

steps until only noise remains, and a network is trained to undo one corruption step at a time; generation then amounts to running the learned chain backwards from pure noise. CSDI makes this chain conditional: the denoising network – a transformer applying attention along time and across assets (Vaswani et al., 2017) – always sees the clean observed entries while denoising the missing ones. Training is self-supervised, hiding a random share of the observed cells and learning to denoise them given the rest. At inference the chain starts from noise on the missing cells, with the observed ones held fixed throughout, and each pass produces one draw from the predictive distribution of the missing block given the observed panel; the final imputation is the median over several draws. CSDI is thus the method from which we explicitly generate multiple conditional draws: the dispersion of its draws is a built-in estimate of the imputation uncertainty, which no other model here exposes. EM also defines a probabilistic conditional model, and could be sampled from in the same way, but we use it only through its conditional mean.

2.3 Evaluation methods

Every experiment produces, for a given missingness pattern and level, an imputed panel $\hat{R} \in \mathbb{R}^{T \times N}$ to be compared against the ground truth R of Section 2.1. Let $\mathcal{M} = \{(t, k) : r_{t,k} \text{ was deleted}\}$ denote the set of imputed cells and $|\mathcal{M}|$ its cardinality. Since our stated goal is not the recovery of individual values but of the panel a downstream model would have been trained on, we score the imputations with two families of metrics. *Pointwise* metrics are computed on the imputed cells alone – the observed cells are returned untouched by every model, and including them would only dilute the error – and measure how close each filled value is to the deleted one. *Structural* metrics instead compare statistics computed from the two full panels, and measure whether the objects that downstream models actually consume (covariances, tail quantiles, marginal distributions, temporal dependence) survive the imputation. The two families need not agree: a method can be nearly optimal pointwise and still destroy the second-moment structure, as zero imputation shows. Deletion is the exception in what follows: it returns a panel with fewer rows, so the pointwise metrics are not defined for it and the structural ones are computed on the surviving subsample.

1. **MAE**: the mean absolute error on the imputed cells,

$$\text{MAE} = \frac{1}{|\mathcal{M}|} \sum_{(t,k) \in \mathcal{M}} |\hat{r}_{t,k} - r_{t,k}|,$$

expressed in return units and robust to occasional large misses.

2. **RMSE**: the root mean squared error on the same cells,

$$\text{RMSE} = \left(\frac{1}{|\mathcal{M}|} \sum_{(t,k) \in \mathcal{M}} (\hat{r}_{t,k} - r_{t,k})^2 \right)^{1/2}.$$

This is the sample counterpart of the corresponding L^2 loss: the imputation minimising its expectation is the conditional expectation given the observed data, so RMSE rewards exactly the conditional means that the likelihood-based models construct, while penalising large errors quadratically.

3. **Covariance error**: the relative Frobenius error of the sample covariance matrix,

$$\text{CovErr} = \frac{\|\hat{\Sigma} - \Sigma\|_F}{\|\Sigma\|_F},$$

where Σ and $\hat{\Sigma}$ are estimated from the true and the imputed panel respectively. This is our central structural metric: the covariance matrix is the structural object this thesis sets out to preserve, and the input of essentially every allocation and risk model built downstream. Being a relative Frobenius error, it weighs the large entries of Σ most, so it is dominated by the variances and by the factor-driven covariances.

4. **Value-at-Risk error**: for each asset the empirical VaR at level $\alpha = 0.05$ is the (sign-flipped) empirical α -quantile of its return series, $\text{VaR}_\alpha(k) = -\hat{q}_\alpha(R_k)$, and we report

$$\text{VaRErr} = \frac{1}{N} \sum_{k=1}^N \left| \widehat{\text{VaR}}_\alpha(k) - \text{VaR}_\alpha(k) \right|.$$

This probes the left tail beyond the second moment, the region that risk management actually regulates – and precisely the region that the stressed pattern of Section 2.1.1 empties out.

5. **Wasserstein error**: the mean 1-Wasserstein distance between the per-asset marginal return distributions,

$$\text{WassErr} = \frac{1}{N} \sum_{k=1}^N W_1(\hat{F}_k, F_k), \quad W_1(F, G) = \int_{\mathbb{R}} |F(x) - G(x)| dx,$$

with F_k and $\hat{F}_k$ the empirical distribution functions of asset k in the true and imputed panel. W_1 measures the minimal average displacement needed to transport one distribution onto the other; it compares the marginals in full, in return units, and detects shape distortions – e.g. imputed values overly concentrated near zero – that moment-based metrics summarise away.

6. **Absolute-return ACF error**: every metric above is invariant to a permutation

of the dates; the last two are not. Daily returns are themselves close to serially uncorrelated, but their absolute values are persistently and positively autocorrelated – volatility clustering, the most robust temporal stylised fact of asset returns (Cont, 2001). Writing $\hat{\rho}_k(\cdot)$ for the sample autocorrelation at lag k , we compare the autocorrelation functions of the absolute returns up to a horizon K ,

$$\text{AbsACFErr} = \frac{1}{NK} \sum_{j=1}^N \sum_{k=1}^K \left| \hat{\rho}_k(|\hat{R}_j|) - \hat{\rho}_k(|R_j|) \right|,$$

with $K = 20$ throughout. An imputer that fills the panel with values that are individually plausible but temporally independent – as any model that treats the dates as exchangeable must – erases this dependence and is caught here and nowhere above.

7. **Leverage error:** negative returns raise subsequent volatility more than positive ones of the same size, the leverage effect (Black, 1976): the cross-correlation curve

$$L_k = \text{corr}(r_t, |r_{t+k}|), \quad k = 1, \dots, K,$$

is negative at short lags. We report the mean absolute gap between the true and imputed curves, averaged over assets and lags as above. Unlike the previous metric, which is blind to sign, this one probes an *asymmetric* dependence between the level of a return and the volatility that follows it, and thereby whether the imputed panel still distinguishes falling from rising markets.

All seven metrics are computed by a single evaluation routine for every model, missingness pattern and missingness level; the resulting tables are collected and discussed in Chapter 3.

2.3.1 Downstream evaluations

The seven metrics above compare the imputed panel with the truth directly. A complementary and in some ways more honest test asks a different question: if we build several models on the imputed panel, how far is their output from each other? We call this class of criteria downstream evaluations. We fix a downstream model, fit it once on each imputed panel, and compare the fitted objects across the imputation methods and against the one fitted on the uncorrupted panel. The aim is to rank the imputation methods against one another rather than against a naive baseline, so we keep these results separate from the tables of Chapter 3. Two such evaluations follow, the optimal portfolio and the factor structure of the panel.

Optimal portfolio

The most classical downstream model in finance is mean-variance portfolio selection (Markowitz, 1952). From a return panel we estimate the mean vector $\hat{\boldsymbol{\mu}}$ and the covariance matrix $\hat{\boldsymbol{\Sigma}}$ and form the fully invested tangency portfolio, the point of the efficient frontier with the highest ratio of expected return to volatility:

$$\hat{w} = \frac{\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\mu}}}{\mathbf{1}^\top \hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\mu}}}, \quad \mathbf{1}^\top \hat{w} = 1.$$

Short positions are allowed, as in the unconstrained classical problem. The covariance enters through its inverse, so any distortion the imputation leaves in $\hat{\boldsymbol{\Sigma}}$, the central structural object evaluated above, is carried into the weights and often amplified there. The portfolio is in this sense a direct stress test of the structural quality that the covariance error of Section 2.3 measures in the aggregate.

In the implementation $\hat{\boldsymbol{\Sigma}}$ is the plain sample covariance of the panel supplied, and $\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\mu}}$ is obtained by an ordinary linear solve. No pseudoinverse, shrinkage or ridge regularisation is applied. For the full-panel imputers this is unproblematic, since $T \gg N$ and the sample covariance is comfortably of full rank. It is not unproblematic for complete-case deletion at the heavier levels, where the surviving subsample has fewer dates than assets: there the sample covariance is singular, no ordinary inverse exists, and the linear solve returns a direction determined by numerical noise in the near-zero eigenvalues. The deletion rows at those levels should accordingly be read as a record of the breakdown, not as meaningful allocations, and we treat them that way in Chapter 3.

We compute $\hat{w}$ once for every imputed panel and once for the uncorrupted panel. To judge an allocation we do not trust the panel it was estimated on: we evaluate each weight vector on the true returns,

$$r_t^p = r_t^\top \hat{w},$$

and report the annualised mean and variance of the resulting series r^p , together with its Sharpe ratio, annualising with 252 trading days and taking a zero risk-free rate. Since the components of r_t are log returns, r_t^p is a weighted log-return series rather than the log return of a self-financing portfolio, which would require weights applied to simple returns and rebalancing; we use it as a linear portfolio-return proxy and nothing in what follows depends on the difference.

Two points about what this exercise does and does not measure. First, the weights and the returns they are evaluated on come from the same period, so the whole comparison is in sample. The portfolio built on the true panel is the oracle: it is the best in-sample allocation available, and no imputation can beat it. What the exercise measures is reconstruction fidelity, how close an imputed panel leads a mean-variance optimiser to the decision the complete panel would have produced, and it is a diagnostic of that and not evidence

about out-of-sample investment performance. Second, we report the distance between the estimated and the oracle weights as $\|\hat{w} - \hat{w}_{\text{True}}\|_1$, which we call the L^1 weight distance; it is a measure of disagreement between two allocations and not the gross exposure of either. A method that preserves the second-moment structure recovers weights near the oracle and comparable realised moments. One that distorts that structure, or that discards data as complete-case deletion does, produces weights that look optimal on the panel it saw and behave poorly on the real returns. The comparison is reported in Chapter 3, in a table of realised moments and a plot of the realised series, kept apart from the metric tables above.

Factor structure

The portfolio reads the covariance through its inverse, so it measures the distortion left by the imputation and the amplification of that distortion at the same time. The second downstream model separates them. We take the statistical factor model of the panel, that is the principal components of $\hat{\Sigma}$, and compare the factors extracted from each imputed panel with those of the uncorrupted one. Nothing is inverted here, so what the comparison reports is the damage to the second moments alone.

Write the spectral decomposition of the estimated covariance as

$$\hat{\Sigma} = \sum_{i=1}^N \hat{\lambda}_i \hat{v}_i \hat{v}_i^\top, \quad \hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_N \geq 0,$$

with orthonormal eigenvectors $\hat{v}_i$. The first k of them, collected in the $N \times k$ matrix $\hat{V}_k$, are the loadings of the k leading statistical factors, and in a panel of equity returns the first of these is the market factor. We report two things. The first is the spectrum, through the share of the total variance

$$\hat{\lambda}_i / \sum_{j=1}^N \hat{\lambda}_j$$

carried by each component, which says whether the imputation has concentrated the panel into fewer factors than the data support or spread it over more. The second is the position of the factor space itself. Let V_k be the true loadings and let $\theta_1 \leq \cdots \leq \theta_k$ be the principal angles between the two subspaces, the cosines of which are the singular values of $V_k^\top \hat{V}_k$. These angles describe how the two k -dimensional subspaces sit relative to one another as a whole. They are ordered alignment angles between the two spans, obtained from a decomposition that recombines the eigenvectors freely, so θ_i is not the error of the i -th principal component and cannot be attached to any individual factor: θ_1 measures the best-aligned direction the two subspaces share and θ_k the worst aligned one, whatever linear combinations of loadings those directions turn out to be. This is exactly what makes the criterion invariant to the sign of each $\hat{v}_i$ and to any reordering among factors of nearly

equal variance, both of which are arbitrary. We summarise the angles by

$$D_k = \frac{1}{\sqrt{k}} \left(\sum_{i=1}^k \sin^2 \theta_i \right)^{1/2} \in [0, 1],$$

the root mean square of their sines, which is 0 when the two factor spaces coincide and 1 when they are orthogonal. Dividing by $\sqrt{k}$ keeps the values comparable across different numbers of factors. D_k is therefore a measure of how well the k -dimensional factor subspace is recovered as a whole, and that is the only reading we give it.

The two downstream evaluations are connected by the same decomposition. Since $\hat{\Sigma}^{-1} = \sum_i \hat{\lambda}_i^{-1} \hat{v}_i \hat{v}_i^\top$, the inverse on which the portfolio of Section 2.3.1 depends weights each direction by the reciprocal of its variance, so it is dominated by the eigenvectors carrying the smallest eigenvalues. These are the directions supported by the least data, and the ones an imputation has the least information to recover. Comparing D_k at a larger k with D_k at a smaller one therefore says how much is lost once the deeper part of the spectrum is included in the subspace to be recovered, which is what the portfolio implicitly relies on. We report D_1 , the recovery of the market direction, and D_5 , the recovery of the five-dimensional leading factor subspace, together with the variance shares, and plot the full spectrum and angle profile, in Chapter 3.

Chapter 3

Results and Discussion

Tables 3.1, 3.2 and 3.3 report the seven evaluation metrics of Section 2.3 for every model, one table per injection pattern of Section 2.1.1. Within each table the levels are ordered by their realised share of missing cells, and within each level the best value of every metric is set in bold. A dash marks a value that is not defined. This happens for the pointwise errors of Deletion, since the panel it returns changes shape, for its remaining metrics at the heaviest levels, where too few complete dates survive, and for KNN at the most extreme stressed level, where the dates to be filled have no well-observed donors left.

The ranking of the models is stable across patterns and levels. The conditional methods dominate the two baselines everywhere, EM is rarely beaten and never by a wide margin, and the neural models match it at a much higher computational cost rather than improving on it; BRITS falls clearly behind at two of the stressed levels.

That EM is so hard to beat is less surprising than it first looks. Daily returns are close to serially uncorrelated, so nearly everything that can be known about a missing entry is carried by the returns observed on the same date, and the conditional mean of Section 2.2 extracts exactly that. What is left over is idiosyncratic, and little of it is predictable from anything else in the panel. The pointwise columns therefore have a floor, and it is reached almost at once: EM’s MAE stays between 0.0071 and 0.0080 across the whole sparse table, and the five conditional methods are separated in the third decimal. The additional capacity of BRITS and CSDI has little to act on, since the expectation of a return given its cross-section is close to linear and the residual is close to noise. They are not punished for that capacity, they simply cannot be paid for it.

The same floor is why the pointwise and the structural columns disagree about severity. MAE and RMSE average over the imputed cells, so they report how hard a typical fill is and stay flat as the number of fills grows: EM’s sparse MAE is 0.0080 at rate 0.001 and 0.0074 at rate 0.3, and zero imputation is just as flat at about 0.011 throughout. The covariance error accumulates those same per-cell errors over the whole panel, and across the same range it grows by a factor of almost forty, from 0.0015 to 0.0565. Read down the two families, the amount of missing data is nearly invisible in the first and is the leading

term in the second. This is the distinction of Section 2.3 between pointwise and structural quality in its plainest form: under the MCAR patterns the MAE of zero imputation is only about fifty percent larger than the best one, while its covariance error is close to an order of magnitude worse than that of EM, so a model can look acceptable entry by entry and still distort the geometry of the panel.

The two families do not even agree on what a good fill is. Minimising squared error asks for a conditional mean, which is by construction less variable than the return it replaces, so the fill that is best entry by entry is also the one that damps the second moment most. The tables show the exchange being made. At the stressed levels (0.005, 0.5, 0.99) and (0.05, 0.5, 0.9) low-rank completion loses to EM on MAE and beats it on the covariance error, and at sparse rate 0.3 BRITS does the same. Low-rank completion is helped here by the norm as much as by the fill: a relative Frobenius error weighs the large entries of Σ , which are the factor-driven ones, so a method that places every fill on the leading subspace reproduces those and neglects the directions the norm barely counts. Section 3.2 shows what that costs.

Which metric is read also decides how much can be told apart at all. The Value-at-Risk and Wasserstein errors are computed one asset at a time, and a marginal distribution of daily returns is easy to reproduce: a fill of roughly the right size leaves it almost intact, which is why both columns are zero to four decimal places at the lighter levels and separate the models only at the heaviest. At sparse rate 0.3 zero imputation is two and a half times worse than the best model on the Value-at-Risk error and nearly nine times worse on the covariance error, from the same panel. The covariance, absolute-return autocorrelation and leverage errors are all built from products of returns, and they require a fill to stand in the right relation to the other assets on its date and to the neighbouring dates. Producing a plausible number is easy; producing a number that is plausible jointly with the rest of the panel is not, and only the second family of metrics can tell the two apart.

The three patterns then differ in which of those relations they destroy. Since the levels are matched on the share of missing cells, the row blocks of the three tables can be read across, and the first result of doing so is that the two MCAR tables are almost interchangeable. EM's covariance error under contiguous gaps is never more than five percent worse than under sparse missingness and at three of the five levels it is better, although the two patterns delete cells in entirely different geometries, isolated entries in one case and runs of up to thirty consecutive days in the other. Breaking the time axis costs a cross-sectional imputer nothing, because on any date the other forty-nine assets are still observed, and for a nearly serially uncorrelated return that is the informative conditioning set anyway. The only model with anything at stake is BRITS, which propagates along time, and the gap table shows both sides of it: at (20, 10) the runs are short, the decayed history is still fresh next to each hole and BRITS has the best MAE and RMSE at that level, while at (2000, 30) that channel has faded and it falls back on the cross-section,

ending up indistinguishable from EM.

The stressed pattern removes the conditioning set itself. Its deletions are drawn by date and are therefore aligned across assets, so the cross-section that the other two patterns leave intact is exactly what thins out, to the point where KNN has no well-observed donors left at the last level and returns nothing at all. At matched shares the gap is wide: at about ten percent missing EM's covariance error is 0.0237 under sparse, 0.0238 under gaps and 0.0554 under stressed, and at about thirty percent it is 0.0565, 0.0589 and 0.6634. The thinner conditioning is the smaller half of the explanation. The deleted returns are not a random draw but the largest ones, so a model fitted to the calm surviving sample fills turbulent dates with calm values and has nothing to tell it otherwise. The error sits in the target rather than in the estimator, which is why the stressed table degrades in a way that no amount of model capacity addresses. At the heaviest level none of the methods tested brings the relative covariance error below 0.6; the turbulent dates are no longer in the data, and the observed sample appears to carry too little about them for any of these models to reconstruct.

This is also what separates the two levels at which BRITS falls behind, (0, 0.1, 0.99) and (0.05, 0.5, 0.9), the two whose deleted entries are the most extreme relative to what stays visible. In the first, every missing cell lies on the one percent most turbulent dates, and BRITS returns an MAE of 0.0301 against EM's 0.0133, behind even KNN. EM's fill is a linear function of the block observed on the same date, so when the other assets move by five percent the fill moves with them, whereas a regression head trained self-supervised on a panel of ordinary dates has learned to produce ordinary values and does not extrapolate. In the second, half of the entries on the most turbulent tenth of dates are gone, which degrades both of BRITS's channels at once, and its covariance error is nearly four times EM's.

The largest numbers in the tables belong to Deletion, and they measure something other than imputation error. It imputes nothing, so its covariance error of 1.4799 at (2000, 30), more than three times that of zero imputation, records how few dates survive the requirement that all fifty assets be observed at once, and under the stressed pattern it records in addition that the survivors are calm by construction.

At the extreme stressed level, finally, the ranking compresses to nothing. Every surviving model reports an MAE between 0.0134 and 0.0137 and a covariance error between 0.648 and 0.663, and zero imputation, hopeless everywhere else, reaches 0.768. The differences between imputation methods are informative only while there is something left in the panel to recover.

Table 3.1: Evaluation scores under sparse missingness (MCAR); levels are the deletion rate r . MAE, RMSE, VaRErr and WassErr are in daily log-return units, CovErr is a relative Frobenius error and AbsACFErr and LevErr are in correlation units; an entry shown as 0.0000 is smaller than 5×10^{-5} .

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
0.001	Deletion	–	–	0.0119	0.0001	0.0001	0.0101	0.0114
	Zero	0.0118	0.0194	0.0026	0.0000	0.0000	0.0009	0.0008
	KNN	0.0099	0.0168	0.0018	0.0000	0.0000	0.0006	0.0005
	LowRank	0.0084	0.0152	0.0016	0.0000	0.0000	0.0005	0.0004
	EM	0.0080	0.0147	0.0015	0.0000	0.0000	0.0005	0.0004
	BRITS	0.0091	0.0158	0.0016	0.0000	0.0000	0.0005	0.0004
	CSDI	0.0081	0.0151	0.0016	0.0000	0.0000	0.0005	0.0004
0.01	Deletion	–	–	0.0532	0.0005	0.0004	0.0247	0.0334
	Zero	0.0107	0.0161	0.0205	0.0001	0.0001	0.0035	0.0027
	KNN	0.0081	0.0123	0.0069	0.0001	0.0000	0.0019	0.0017
	LowRank	0.0077	0.0117	0.0054	0.0000	0.0000	0.0016	0.0015
	EM	0.0071	0.0109	0.0051	0.0000	0.0000	0.0015	0.0014
	BRITS	0.0076	0.0116	0.0064	0.0001	0.0000	0.0017	0.0016
	CSDI	0.0069	0.0107	0.0050	0.0000	0.0000	0.0015	0.0014
0.05	Deletion	–	–	0.1951	0.0019	0.0012	0.1334	0.0515
	Zero	0.0111	0.0167	0.1044	0.0005	0.0006	0.0132	0.0073
	KNN	0.0084	0.0127	0.0283	0.0003	0.0002	0.0047	0.0045
	LowRank	0.0078	0.0118	0.0169	0.0002	0.0001	0.0045	0.0041
	EM	0.0072	0.0113	0.0151	0.0002	0.0002	0.0044	0.0039
	BRITS	0.0073	0.0113	0.0145	0.0002	0.0002	0.0043	0.0040
	CSDI	0.0072	0.0112	0.0149	0.0002	0.0002	0.0051	0.0041
0.1	Deletion	–	–	0.4492	0.0091	0.0051	–	–
	Zero	0.0109	0.0163	0.1872	0.0011	0.0011	0.0276	0.0111
	KNN	0.0083	0.0125	0.0487	0.0007	0.0004	0.0090	0.0067
	LowRank	0.0077	0.0116	0.0282	0.0004	0.0003	0.0072	0.0058
	EM	0.0071	0.0110	0.0237	0.0005	0.0003	0.0068	0.0056
	BRITS	0.0072	0.0111	0.0238	0.0005	0.0003	0.0068	0.0057
	CSDI	0.0072	0.0111	0.0261	0.0004	0.0003	0.0077	0.0057
0.3	Deletion	–	–	–	–	–	–	–
	Zero	0.0109	0.0161	0.4749	0.0037	0.0033	0.0638	0.0201
	KNN	0.0087	0.0129	0.1467	0.0024	0.0014	0.0173	0.0118
	LowRank	0.0080	0.0119	0.0749	0.0014	0.0007	0.0170	0.0105
	EM	0.0074	0.0112	0.0565	0.0017	0.0009	0.0150	0.0099
	BRITS	0.0076	0.0114	0.0536	0.0015	0.0008	0.0149	0.0101
	CSDI	0.0074	0.0113	0.0641	0.0014	0.0008	0.0203	0.0112

Table 3.2: Evaluation scores under contiguous-gap missingness (MCAR); levels are the number of seeds and the mean run length (n, ℓ) of Section 2.1.1. MAE, RMSE, VaRErr and WassErr are in daily log-return units, CovErr is a relative Frobenius error and AbsACFErr and LevErr are in correlation units; an entry shown as 0.0000 is smaller than 5×10^{-5} .

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(20, 10)	Deletion	–	–	0.0103	0.0001	0.0001	0.0033	0.0033
	Zero	0.0121	0.0184	0.0027	0.0000	0.0000	0.0003	0.0002
	KNN	0.0096	0.0144	0.0016	0.0000	0.0000	0.0002	0.0001
	LowRank	0.0087	0.0123	0.0010	0.0000	0.0000	0.0002	0.0001
	EM	0.0079	0.0117	0.0011	0.0000	0.0000	0.0001	0.0001
	BRITS	0.0078	0.0112	0.0010	0.0000	0.0000	0.0001	0.0001
	CSDI	0.0079	0.0115	0.0009	0.0000	0.0000	0.0001	0.0001
(200, 10)	Deletion	–	–	0.1492	0.0010	0.0004	0.0243	0.0158
	Zero	0.0107	0.0156	0.0222	0.0001	0.0001	0.0037	0.0015
	KNN	0.0082	0.0121	0.0065	0.0001	0.0000	0.0016	0.0012
	LowRank	0.0078	0.0118	0.0061	0.0001	0.0000	0.0015	0.0012
	EM	0.0072	0.0109	0.0046	0.0001	0.0000	0.0014	0.0011
	BRITS	0.0073	0.0111	0.0049	0.0001	0.0000	0.0015	0.0011
	CSDI	0.0072	0.0109	0.0049	0.0001	0.0000	0.0015	0.0011
(1000, 10)	Deletion	–	–	0.1825	0.0017	0.0010	0.1135	0.0493
	Zero	0.0104	0.0152	0.0785	0.0004	0.0005	0.0155	0.0048
	KNN	0.0080	0.0118	0.0224	0.0003	0.0002	0.0059	0.0032
	LowRank	0.0074	0.0108	0.0137	0.0002	0.0001	0.0048	0.0029
	EM	0.0069	0.0103	0.0116	0.0002	0.0001	0.0049	0.0028
	BRITS	0.0070	0.0104	0.0116	0.0002	0.0001	0.0049	0.0028
	CSDI	0.0069	0.0103	0.0114	0.0002	0.0001	0.0051	0.0027
(1000, 20)	Deletion	–	–	0.8524	0.0087	0.0042	0.1872	0.1681
	Zero	0.0110	0.0163	0.1716	0.0010	0.0010	0.0354	0.0068
	KNN	0.0085	0.0127	0.0487	0.0007	0.0004	0.0117	0.0049
	LowRank	0.0078	0.0119	0.0310	0.0004	0.0003	0.0088	0.0046
	EM	0.0073	0.0114	0.0238	0.0005	0.0003	0.0083	0.0044
	BRITS	0.0074	0.0114	0.0241	0.0005	0.0003	0.0082	0.0045
	CSDI	0.0073	0.0115	0.0328	0.0004	0.0003	0.0095	0.0044
(2000, 30)	Deletion	–	–	1.4799	0.0104	0.0100	–	–
	Zero	0.0108	0.0161	0.4324	0.0028	0.0027	0.0908	0.0143
	KNN	0.0085	0.0128	0.1378	0.0018	0.0011	0.0277	0.0090
	LowRank	0.0080	0.0120	0.0820	0.0010	0.0006	0.0218	0.0083
	EM	0.0073	0.0113	0.0589	0.0012	0.0007	0.0212	0.0079
	BRITS	0.0074	0.0114	0.0555	0.0011	0.0006	0.0187	0.0082
	CSDI	0.0074	0.0115	0.0896	0.0010	0.0007	0.0265	0.0091

Table 3.3: Evaluation scores under stressed missingness (MNAR); levels are the parameters (r, h, p) of Section 2.1.1. MAE, RMSE, VaRErr and WassErr are in daily log-return units, CovErr is a relative Frobenius error and AbsACFErr and LevErr are in correlation units; an entry shown as 0.0000 is smaller than 5×10^{-5} .

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(0, 0.1, 0.99)	Deletion	–	–	0.2813	0.0009	0.0004	0.0582	0.0200
	Zero	0.0552	0.0640	0.0676	0.0001	0.0001	0.0115	0.0070
	KNN	0.0202	0.0285	0.0229	0.0000	0.0000	0.0034	0.0029
	LowRank	0.0158	0.0220	0.0116	0.0000	0.0000	0.0025	0.0020
	EM	0.0133	0.0187	0.0094	0.0000	0.0000	0.0021	0.0017
	BRITS	0.0301	0.0385	0.0253	0.0000	0.0000	0.0046	0.0036
	CSDI	0.0163	0.0226	0.0175	0.0000	0.0000	0.0027	0.0022
(0.005, 0.5, 0.99)	Deletion	–	–	0.2960	0.0010	0.0005	0.0687	0.0268
	Zero	0.0321	0.0463	0.2238	0.0005	0.0003	0.0483	0.0172
	KNN	0.0161	0.0261	0.0991	0.0001	0.0001	0.0168	0.0095
	LowRank	0.0129	0.0200	0.0279	0.0000	0.0001	0.0078	0.0067
	EM	0.0118	0.0189	0.0319	0.0000	0.0001	0.0077	0.0065
	BRITS	0.0122	0.0197	0.0401	0.0000	0.0001	0.0082	0.0069
	CSDI	0.0136	0.0233	0.0751	0.0000	0.0001	0.0132	0.0087
(0.04, 0.2, 0.95)	Deletion	–	–	0.5325	0.0043	0.0018	0.1079	0.0491
	Zero	0.0143	0.0219	0.2254	0.0010	0.0007	0.0375	0.0143
	KNN	0.0092	0.0141	0.0562	0.0003	0.0002	0.0092	0.0067
	LowRank	0.0083	0.0125	0.0253	0.0002	0.0001	0.0068	0.0052
	EM	0.0077	0.0117	0.0217	0.0002	0.0001	0.0062	0.0048
	BRITS	0.0078	0.0118	0.0231	0.0002	0.0001	0.0064	0.0049
	CSDI	0.0078	0.0122	0.0374	0.0002	0.0001	0.0078	0.0061
(0.05, 0.5, 0.9)	Deletion	–	–	0.6318	0.0056	0.0023	0.1230	0.0630
	Zero	0.0174	0.0249	0.5125	0.0030	0.0017	0.0843	0.0244
	KNN	0.0103	0.0161	0.1602	0.0010	0.0004	0.0258	0.0124
	LowRank	0.0091	0.0139	0.0483	0.0004	0.0002	0.0110	0.0091
	EM	0.0084	0.0130	0.0554	0.0008	0.0003	0.0107	0.0087
	BRITS	0.0106	0.0162	0.2083	0.0015	0.0006	0.0256	0.0126
	CSDI	0.0086	0.0137	0.0542	0.0008	0.0003	0.0165	0.0109
(0.2, 1, 0.9)	Deletion	–	–	–	–	–	–	–
	Zero	0.0149	0.0219	0.7677	0.0070	0.0042	0.1201	0.0333
	KNN	–	–	–	–	–	–	–
	LowRank	0.0137	0.0212	0.6525	0.0062	0.0029	0.1111	0.0316
	EM	0.0134	0.0210	0.6634	0.0065	0.0031	0.1112	0.0315
	BRITS	0.0136	0.0214	0.6480	0.0063	0.0025	0.1069	0.0299
	CSDI	0.0134	0.0210	0.6559	0.0064	0.0028	0.1065	0.0298

3.1 Downstream evaluation: the optimal portfolio

We now turn to the downstream evaluation of Section 2.3.1. For every imputed panel, and for the uncorrupted one, we estimate the tangency portfolio and evaluate it on the true returns. Tables 3.4, 3.5 and 3.6 report the annualised mean, variance and Sharpe ratio of the resulting weighted log-return series, together with the L^1 weight distance $\|\Delta w\|_1$ between the estimated weights and the oracle's. The oracle is the same in every row, with an annualised mean of 0.361, a variance of 0.049 and a Sharpe ratio of 1.636.

The exercise is in sample throughout: the weights and the returns they are evaluated on come from the same period, so the oracle is the best allocation available and no imputation can beat its Sharpe. What the tables measure is reconstruction fidelity, how close an imputed panel brings a mean-variance optimiser to the decision the complete panel would have supported, and they should not be read as evidence about out-of-sample investment performance. The two naive baselines are kept out of the tables and discussed at the end, so that the comparison stays among the imputation methods themselves.

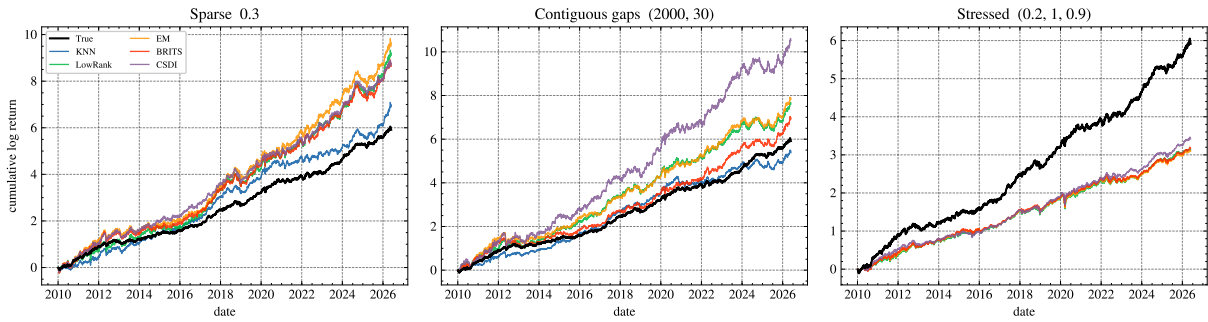


Figure 3.1: Cumulative weighted log return of the tangency portfolio at the heaviest level of each missingness pattern. The oracle, estimated on the uncorrupted panel, is drawn in black; the five imputers are estimated on the corrupted panels and evaluated on the same true returns, so the exercise is in sample throughout and the series are linear portfolio-return proxies rather than the log returns of self-financing portfolios. Under the two MCAR patterns (left, centre) the imputed portfolios drift above the oracle by taking on more risk, while under the stressed pattern (right) they fall short of it. The two baselines, off the vertical scale, are omitted.

At the lightest levels there is nothing to separate them. Under sparse missingness at rate 0.001, and under the shortest gaps, every imputer reproduces the oracle Sharpe to within a thousandth and moves the weights by an L^1 distance of less than 0.35. The gap opens only as the missingness grows, and even then the five methods stay close to one another, with no single model leading throughout: CSDI is marginally ahead on the light sparse levels, EM and low-rank completion on the gaps, EM again on most of the stressed levels, and the best entries at the easy levels are little more than ties. What separates the methods is less who leads than who fails. EM and low-rank completion are the steadiest, never the worst at any level. BRITS keeps pace until the stressed level (0.05, 0.5, 0.9),

Table 3.4: Optimal-portfolio evaluation under sparse missingness. Annualised mean, variance and Sharpe ratio of the weighted log-return series obtained by applying to the true returns the tangency weights estimated from each panel, together with the L^1 weight distance $\|\hat{w} - \hat{w}_{\text{True}}\|_1$, and the oracle estimated on the uncorrupted data. The comparison is in sample and diagnoses reconstruction fidelity, not investment performance.

Level	Model	Mean	Variance	Sharpe	$\ \Delta w\ _1$
0.001	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.362	0.0491	1.635	0.30
	LowRank	0.362	0.0490	1.635	0.29
	EM	0.362	0.0491	1.634	0.30
	BRITS	0.362	0.0492	1.634	0.33
	CSDI	0.362	0.0489	1.635	0.30
0.01	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.358	0.0486	1.623	1.11
	LowRank	0.360	0.0492	1.624	1.08
	EM	0.361	0.0493	1.625	1.03
	BRITS	0.359	0.0487	1.625	0.94
	CSDI	0.361	0.0492	1.626	1.02
0.05	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.373	0.0564	1.569	2.92
	LowRank	0.391	0.0625	1.565	3.01
	EM	0.391	0.0623	1.567	3.06
	BRITS	0.384	0.0597	1.572	2.81
	CSDI	0.396	0.0627	1.580	2.89
0.1	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.351	0.0554	1.490	4.07
	LowRank	0.375	0.0603	1.529	3.33
	EM	0.377	0.0607	1.531	3.40
	BRITS	0.371	0.0577	1.542	3.10
	CSDI	0.373	0.0597	1.527	3.03
0.3	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.422	0.1086	1.279	8.43
	LowRank	0.556	0.1970	1.253	12.21
	EM	0.581	0.2130	1.259	13.76
	BRITS	0.528	0.1852	1.226	12.61
	CSDI	0.531	0.1723	1.279	11.33

Table 3.5: Optimal-portfolio evaluation under contiguous-gap missingness. Annualised mean, variance and Sharpe ratio of the weighted log-return series obtained by applying to the true returns the tangency weights estimated from each panel, together with the L^1 weight distance $\|\hat{w} - \hat{w}_{\text{True}}\|_1$, and the oracle estimated on the uncorrupted data. The comparison is in sample and diagnoses reconstruction fidelity, not investment performance.

Level	Model	Mean	Variance	Sharpe	$\ \Delta w\ _1$
(20, 10)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.363	0.0492	1.635	0.16
	LowRank	0.362	0.0491	1.635	0.11
	EM	0.362	0.0491	1.635	0.10
	BRITS	0.362	0.0490	1.635	0.10
	CSDI	0.362	0.0489	1.635	0.11
(200, 10)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.363	0.0497	1.627	0.87
	LowRank	0.365	0.0503	1.627	0.84
	EM	0.363	0.0499	1.628	0.84
	BRITS	0.362	0.0494	1.629	0.73
	CSDI	0.364	0.0499	1.628	0.79
(1000, 10)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.372	0.0557	1.575	2.24
	LowRank	0.369	0.0537	1.593	1.91
	EM	0.371	0.0539	1.596	1.77
	BRITS	0.369	0.0540	1.588	1.99
	CSDI	0.374	0.0555	1.589	1.97
(1000, 20)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.380	0.0644	1.500	4.31
	LowRank	0.386	0.0612	1.560	3.14
	EM	0.379	0.0598	1.549	3.40
	BRITS	0.358	0.0529	1.555	3.20
	CSDI	0.381	0.0611	1.542	3.52
(2000, 30)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.329	0.0767	1.186	6.52
	LowRank	0.465	0.1439	1.225	9.14
	EM	0.478	0.1460	1.250	10.17
	BRITS	0.423	0.1098	1.276	7.50
	CSDI	0.643	0.2937	1.186	14.48

Table 3.6: Optimal-portfolio evaluation under stressed missingness. Annualised mean, variance and Sharpe ratio of the weighted log-return series obtained by applying to the true returns the tangency weights estimated from each panel, together with the L^1 weight distance $\|\hat{w} - \hat{w}_{\text{True}}\|_1$, and the oracle estimated on the uncorrupted data. The comparison is in sample and diagnoses reconstruction fidelity, not investment performance.

Level	Model	Mean	Variance	Sharpe	$\ \Delta w\ _1$
(0, 0.1, 0.99)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.347	0.0459	1.619	1.13
	LowRank	0.356	0.0477	1.629	0.76
	EM	0.356	0.0478	1.631	0.60
	BRITS	0.345	0.0452	1.620	1.20
	CSDI	0.360	0.0487	1.630	0.66
(0.005, 0.5, 0.99)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.328	0.0439	1.565	2.50
	LowRank	0.353	0.0483	1.606	1.58
	EM	0.353	0.0480	1.612	1.44
	BRITS	0.349	0.0476	1.602	1.78
	CSDI	0.375	0.0572	1.566	2.91
(0.04, 0.2, 0.95)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.359	0.0543	1.542	3.16
	LowRank	0.365	0.0529	1.586	2.09
	EM	0.370	0.0545	1.587	2.00
	BRITS	0.369	0.0536	1.593	1.91
	CSDI	0.392	0.0615	1.580	2.62
(0.05, 0.5, 0.9)	True (oracle)	0.361	0.0488	1.636	–
	KNN	0.346	0.0597	1.418	4.64
	LowRank	0.359	0.0576	1.494	3.50
	EM	0.359	0.0568	1.506	3.45
	BRITS	0.243	0.0347	1.304	4.49
	CSDI	0.354	0.0551	1.508	3.50
(0.2, 1, 0.9)	True (oracle)	0.361	0.0488	1.636	–
	KNN	–	–	–	–
	LowRank	0.191	0.0255	1.195	5.30
	EM	0.189	0.0245	1.207	5.22
	BRITS	0.193	0.0255	1.212	5.43
	CSDI	0.209	0.0266	1.281	5.22

where it turns too cautious and its Sharpe falls to 1.30, against the 1.51 of EM. KNN holds the tightest weights under heavy sparse missingness but is left without donors at the most extreme stressed level and cannot be computed there. CSDI is the closest of all at that same extreme level, and yet at the heaviest gap it takes on the most risk of any method, with a realised variance of 0.29 and a weight distance above 14.

That last case is an instance of the general pattern in how the portfolios degrade. As the missingness grows the imputed panels do not lose a little Sharpe gracefully; their realised variance climbs well above the oracle's, from 0.049 to values past 0.2 at the heaviest sparse and gap levels, and the weights swell to many times their original size. This is the error maximisation that mean-variance optimisation is known for. A covariance matrix that the imputation has distorted, once inverted, produces an over-confident portfolio that takes large offsetting positions the true second moments never supported. The realised mean rises together with the variance, so in a market that trended upward over the sample these aggressive portfolios finish higher in level than the oracle while carrying far more risk, which the Sharpe ratio correctly discounts.

Figure 3.1 makes the two regimes visible at the heaviest level of each pattern. Under the two MCAR patterns the imputed portfolios end above the oracle, with visibly larger swings along the way: the distorted covariance inflates risk, and in this sample the extra risk happened to pay. The stressed pattern is the mirror image. There the missing dates are the turbulent ones, the imputers never see the volatility they would need to price, and their portfolios fall short of the oracle outright, ending at little more than half of its cumulative return. CSDI comes closest, but the shortfall that none of them close is the one the covariance error already exposed in Section 2.3: under MNAR the information needed to rebuild the portfolio is missing from the observed panel, and no imputer can supply it.

The baselines show, from below, how much the imputation matters once a model is built on top of it. Zero imputation trails the conditional methods at almost every level, the exception being the very heaviest, where its heavy damping of the covariance, a liability everywhere else, accidentally restrains the over-leverage that afflicts the others. Complete-case deletion is worse, and in a way the reconstruction metrics could only suggest. Estimating the covariance from the surviving dates alone leaves it so far from the truth that the optimiser levers into unstable portfolios: the realised variance is already more than three times the oracle's at sparse rate 0.05, the Sharpe falls to almost zero at rate 0.1, and at the gap level (1000, 20) the portfolio's mean return turns negative. These last figures should be read with the caveat of Section 2.3.1: by then the complete-case sample has fewer dates than assets, so its covariance matrix is singular, no ordinary inverse exists, and the weights are fixed by numerical noise in the near-zero eigenvalues rather than by any estimate the data supports. They record the breakdown rather than measure it. At the heaviest levels deletion cannot be evaluated at all, too few complete dates surviving to

attempt a covariance. A method that reads as merely inefficient in the metric tables can, once it feeds a portfolio, produce an allocation that is dangerously unstable.

3.2 Downstream evaluation: the factor structure

The second downstream model is the one of Section 2.3.1. From every imputed panel, and from the uncorrupted one, we extract the principal components of the estimated covariance and compare the resulting factor model with the true one. Tables 3.7, 3.8 and 3.9 report the share of the panel variance carried by the first component and by the first five, together with the subspace distances D_1 and D_5 . On the uncorrupted panel the first principal component carries 41.74% of the variance and the first five carry 60.02%, and these are the values every row is read against. As with the portfolio, the two naive baselines are kept out of the tables and discussed at the end, so that the comparison stays among the imputation methods themselves.

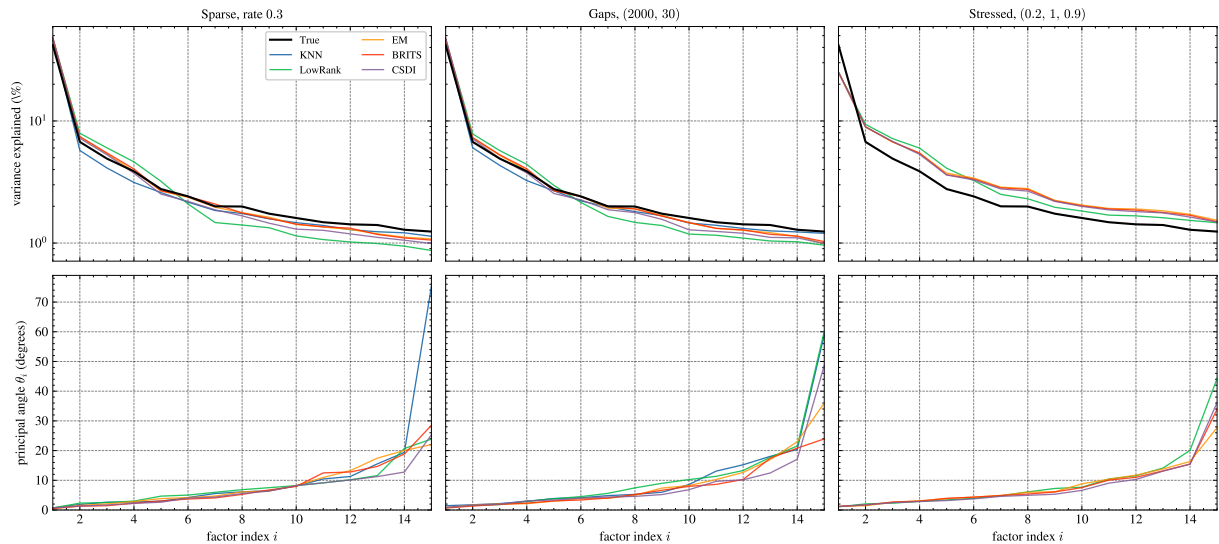


Figure 3.2: Factor structure at the heaviest level of each missingness pattern. Top: share of the panel variance carried by each of the first fifteen principal components, on a logarithmic scale, with the uncorrupted panel in black. Bottom: the fifteen principal angles between the true and the estimated fifteen-factor subspaces, in increasing order; here the horizontal index enumerates the ordered principal angles and not the individual factors, so θ_i should not be read as the error of the i -th principal component. The best-aligned directions the two subspaces share are recovered to within a few degrees by every method, while the worst-aligned ones are not.

The first factor survives almost everything. Under sparse missingness at rate 0.3, where a third of the panel is gone, D_1 stays below 0.026 for every method, an angle of less than a degree and a half between the estimated market factor and the true one. The five-factor space is a different matter: at the same level D_5 runs from 0.14 to 0.39, between six and eighteen times larger. Widening the subspace from one dimension to five therefore costs

Table 3.7: Factor-structure evaluation under sparse missingness. Share of the panel variance carried by the first principal component and by the first five, and normalised subspace distance between the estimated and the true leading eigenspaces at one and five factors. D_1 and D_5 measure how well each subspace is recovered as a whole and are not per-factor errors.

Level	Model	λ_1 (%)	Top-5 (%)	D_1	D_5
0.001	True (reference)	41.74	60.02	–	–
	KNN	41.76	60.04	0.0006	0.0051
	LowRank	41.77	60.06	0.0006	0.0036
	EM	41.77	60.06	0.0005	0.0040
	BRITS	41.77	60.05	0.0005	0.0046
	CSDI	41.77	60.06	0.0006	0.0041
0.01	True (reference)	41.74	60.02	–	–
	KNN	41.87	60.09	0.0030	0.0143
	LowRank	41.91	60.26	0.0026	0.0195
	EM	41.91	60.22	0.0026	0.0126
	BRITS	41.87	60.16	0.0034	0.0154
	CSDI	41.91	60.22	0.0027	0.0126
0.05	True (reference)	41.74	60.02	–	–
	KNN	42.28	60.32	0.0082	0.0364
	LowRank	42.64	61.37	0.0068	0.0809
	EM	42.64	61.19	0.0062	0.0440
	BRITS	42.59	61.14	0.0054	0.0347
	CSDI	42.72	61.18	0.0062	0.0323
0.1	True (reference)	41.74	60.02	–	–
	KNN	43.00	60.35	0.0133	0.0436
	LowRank	43.60	62.56	0.0100	0.2551
	EM	43.59	62.05	0.0094	0.1120
	BRITS	43.47	61.96	0.0100	0.1034
	CSDI	43.73	62.03	0.0122	0.1014
0.3	True (reference)	41.74	60.02	–	–
	KNN	45.62	61.19	0.0252	0.1434
	LowRank	47.56	69.35	0.0215	0.3914
	EM	47.88	67.16	0.0152	0.2039
	BRITS	47.09	66.74	0.0161	0.2376
	CSDI	48.54	67.28	0.0211	0.1954

Table 3.8: Factor-structure evaluation under contiguous-gap missingness. Share of the panel variance carried by the first principal component and by the first five, and normalised subspace distance between the estimated and the true leading eigenspaces at one and five factors. D_1 and D_5 measure how well each subspace is recovered as a whole and are not per-factor errors.

Level	Model	λ_1 (%)	Top-5 (%)	D_1	D_5
(20, 10)	True (reference)	41.74	60.02	–	–
	KNN	41.75	60.02	0.0008	0.0040
	LowRank	41.75	60.03	0.0004	0.0033
	EM	41.75	60.03	0.0005	0.0034
	BRITS	41.75	60.03	0.0004	0.0029
	CSDI	41.75	60.03	0.0003	0.0027
(200, 10)	True (reference)	41.74	60.02	–	–
	KNN	41.90	60.15	0.0034	0.0117
	LowRank	41.93	60.26	0.0031	0.0207
	EM	41.92	60.24	0.0021	0.0117
	BRITS	41.92	60.24	0.0024	0.0115
	CSDI	41.92	60.24	0.0028	0.0100
(1000, 10)	True (reference)	41.74	60.02	–	–
	KNN	42.29	60.21	0.0089	0.0376
	LowRank	42.42	61.07	0.0053	0.0334
	EM	42.42	60.88	0.0047	0.0249
	BRITS	42.37	60.84	0.0046	0.0270
	CSDI	42.43	60.85	0.0040	0.0291
(1000, 20)	True (reference)	41.74	60.02	–	–
	KNN	42.93	60.62	0.0179	0.0525
	LowRank	43.63	62.50	0.0143	0.1892
	EM	43.52	62.14	0.0097	0.0550
	BRITS	43.44	62.08	0.0104	0.0508
	CSDI	43.78	62.22	0.0158	0.0453
(2000, 30)	True (reference)	41.74	60.02	–	–
	KNN	44.63	60.92	0.0511	0.1365
	LowRank	46.84	67.76	0.0375	0.3291
	EM	46.95	66.09	0.0262	0.1273
	BRITS	46.36	65.70	0.0252	0.1177
	CSDI	47.94	66.31	0.0465	0.1435

Table 3.9: Factor-structure evaluation under stressed missingness. Share of the panel variance carried by the first principal component and by the first five, and normalised subspace distance between the estimated and the true leading eigenspaces at one and five factors. D_1 and D_5 measure how well each subspace is recovered as a whole and are not per-factor errors.

Level	Model	λ_1 (%)	Top-5 (%)	D_1	D_5
(0, 0.1, 0.99)	True (reference)	41.74	60.02	–	–
	KNN	41.31	59.70	0.0106	0.0276
	LowRank	41.75	60.12	0.0079	0.0152
	EM	41.73	60.07	0.0063	0.0145
	BRITS	41.45	59.73	0.0176	0.0317
	CSDI	42.06	60.23	0.0088	0.0160
(0.005, 0.5, 0.99)	True (reference)	41.74	60.02	–	–
	KNN	39.81	58.28	0.0309	0.1055
	LowRank	41.92	60.67	0.0180	0.0555
	EM	41.80	60.14	0.0178	0.0950
	BRITS	41.56	59.92	0.0203	0.0900
	CSDI	43.62	61.24	0.0316	0.1166
(0.04, 0.2, 0.95)	True (reference)	41.74	60.02	–	–
	KNN	41.68	59.49	0.0169	0.0680
	LowRank	42.73	61.36	0.0149	0.0714
	EM	42.66	61.03	0.0126	0.0327
	BRITS	42.58	60.96	0.0132	0.0368
	CSDI	43.27	61.34	0.0170	0.0383
(0.05, 0.5, 0.9)	True (reference)	41.74	60.02	–	–
	KNN	40.27	57.71	0.0398	0.1131
	LowRank	43.50	63.34	0.0268	0.3293
	EM	43.29	61.77	0.0232	0.0843
	BRITS	40.36	59.06	0.1407	0.1219
	CSDI	44.63	62.49	0.0303	0.0964
(0.2, 1, 0.9)	True (reference)	41.74	60.02	–	–
	KNN	–	–	–	–
	LowRank	25.12	51.75	0.1186	0.3814
	EM	24.81	49.58	0.1148	0.1343
	BRITS	25.19	50.00	0.1106	0.1184
	CSDI	25.01	49.66	0.1141	0.1340

a great deal, and the misalignment within it is concentrated in a single direction rather than shared out evenly. At sparse rate 0.3 the four smallest principal angles between the true and the estimated five-factor subspaces stay within about six degrees for KNN, EM, BRITS and CSDI, while the largest of the five runs between 16.5 and 31.4 degrees. What this locates is a direction inside the leading five-dimensional space that the imputations fail to place, not a defect of the fifth principal component as such. A plausible reason is that the spectrum stops separating the components there: the fifth and sixth true eigenvalues carry 2.77% and 2.41% of the variance, close enough that little in the data pins down the plane they span, and a subspace cut through directions that the data barely distinguishes is the part hardest to recover. The angle profiles in Figure 3.2 show the same shape over a longer range: the smaller principal angles stay small and the profile climbs steeply only in its last few entries.

The spectrum is distorted in a consistent direction. Under the tested MCAR configurations every conditional imputer raises the weight of the first factor, from 41.74% to between 45.6% and 48.5% at sparse rate 0.3, and raises the five-factor share from 60.02% to between 61.2% and 69.3%. This is what point imputation would lead one to expect. Each filled value is a conditional location estimate formed from the assets observed on the same date, so it carries less idiosyncratic variation than a real return and lies closer to the dominant directions of the panel, and the more entries are filled the more the panel comes to look like its own factor model. The effect is a property of inserting point estimates rather than draws, and we observe it here for these methods under these patterns; a multiple-imputation treatment that preserved the predictive spread would not be expected to show it in the same form. The top row of Figure 3.2 shows the effect across the whole spectrum: under the two MCAR patterns the imputed curves sit above the true one at the head and below it in the tail. A downstream user reading these panels would conclude that the market is more tightly coupled than it is, and would underestimate the idiosyncratic variance that a hedge has to absorb.

Among the methods, low-rank completion behaves in a way worth isolating. It has the largest D_5 at ten of the fifteen levels, and at eight of the nine heaviest, reaching 0.39 under sparse missingness at rate 0.3 and 0.33 at the stressed level (0.05, 0.5, 0.9), against 0.21 and 0.08 for EM at the same levels. Its five-factor variance share is also the most inflated, 69.3% against a true 60.0%. The direction of this result is worth stating plainly, because it is the opposite of what the method’s construction would suggest. Low-rank completion is the one imputer that explicitly imposes a factor structure, and an evaluation of factor structure might be expected to favour it. What happens instead is that truncating the spectrum removes precisely the weakly separated middle directions: at sparse rate 0.3 the largest principal angle between its five-factor subspace and the true one is 55.8 degrees, where the worst of the other methods reaches 31.4. It recovers the market direction as well as anyone and pays for it deeper into the subspace.

BRITS fails once, and the failure is informative. At the stressed level (0.05, 0.5, 0.9) its D_1 is 0.141, an angle of eight degrees between the estimated and the true market direction, where the other four methods stay between 0.023 and 0.040, under two and a half degrees. This is the same level at which its portfolio Sharpe collapsed to 1.30 in Table 3.6 while EM and CSDI held above 1.50. The portfolio evaluation registered that something had gone wrong but could not say what; the factor evaluation locates it, in a rotation of the market direction itself, which no amount of care further down the spectrum can compensate.

At the most extreme stressed level the ranking dissolves and all the methods fail together. The first factor falls from 41.74% to about 25% and the five-factor share from 60.02% to about 50%, for low-rank completion, EM, BRITS and CSDI alike, with D_1 between 0.111 and 0.119 for all of them and KNN left without donors. The mechanism explains the uniformity. Stressed missingness removes the most turbulent dates, and those are the dates on which correlations rise and the market factor dominates, so what has been deleted is not a random part of the panel but the part that carries the factor structure. The right-hand column of Figure 3.2 shows every imputed spectrum lying flatter than the truth. This is the same conclusion the portfolio reached at this level, and the same one the covariance error of Section 2.3 reached before it: under this stressed pattern the information appears not to be recoverable from the observed panel, and none of the methods tested here separate.

The baselines fail in opposite directions, which is itself instructive. Zero imputation dilutes the factor structure rather than concentrating it, taking the first component down to 30.6% at sparse rate 0.3 and to 18.1% at the extreme stressed level, since a constant inserted in place of a return is uncorrelated with everything else in the cross section. Complete-case deletion is erratic in a way the reconstruction metrics never suggested. At the gap level (1000, 20) only 33 dates survive and its leading eigenvector is nearly orthogonal to the true one, $D_1 = 0.977$; at (2000, 30) three dates survive and five components exhaust the variance of the panel entirely; at sparse rate 0.3 and at the extreme stressed level no complete date survives at all and nothing can be estimated. Between them the two baselines bracket the failure: one flattens the factor structure, the other invents one.

The two downstream evaluations meet here. Since $\hat{\Sigma}^{-1} = \sum_i \hat{\lambda}_i^{-1} \hat{v}_i \hat{v}_i^\top$, the inverse the portfolio depends on weights each direction by the reciprocal of its variance, so it loads most heavily on the trailing eigenvectors, which are exactly the directions the angle profiles show to be least reliable. The error maximisation of Section 3.1 is therefore not a separate defect of mean-variance optimisation. It is the distortion measured here, seen through an inverse that magnifies whatever the imputation has left in the parts of the spectrum it understands worst.

3.3 Computational costs

The scores above say nothing about what each model costs to produce, so we close the chapter with the computational side of the experiments. Everything in this thesis, training included, was run on a single consumer machine, an Apple Mac mini with the M4 chip (four performance and six efficiency cores) and 16 GB of unified memory. The neural models run on the CPU build of PyTorch, so no GPU acceleration is involved. Both networks are deliberately small, with roughly 1.2×10^5 trainable parameters for BRITS and 4.7×10^4 for CSDI, which means that the costs reported here come from the algorithms themselves and not from the size of the models.

The classical methods have no training stage and a negligible running time: the whole family fills a 4125×50 panel in a few seconds, and EM, the most expensive of the group with its fifty covariance sweeps, needs about one second. The neural models are on another scale. Training on a single corrupted panel took just under an hour for BRITS (10^4 epochs) and just under four hours for CSDI (2000 epochs), so covering the fifteen levels of Section 2.1.1 required roughly seventy hours of computation, four fifths of which were spent on CSDI. CSDI is by far the most expensive model at imputation time as well. BRITS fills the panel with a single forward pass in a fraction of a second, while a CSDI imputation is the median of 32 stochastic draws, each of which runs the full 50-step reverse diffusion over every window of the panel. This takes roughly twenty-four minutes per panel, and the evaluation sweep behind Tables 3.1, 3.2 and 3.3 ran for about six hours, almost all of them spent sampling from CSDI. In view of the tables above, this cost buys very little: EM reaches the same quality for about one second per panel, four orders of magnitude less than CSDI, which rarely beats it.

Chapter 4

Conclusion

4.1 Summary

This thesis measured what is lost when the missing entries of a panel of daily equity returns are filled in. Fifty S&P 500 constituents over sixteen years were corrupted under three mechanisms at five severities each, reconstructed by seven methods, and scored both against the truth directly and through two models a user would build on the result. We return to the four questions of Section 1.3.

In this experiment the mechanism mattered more than the amount. Under the two random patterns the methods hold up even when a third of the panel is gone: at sparse rate 0.3 the market direction is still recovered to within a degree and a half of the true one. Under the stressed pattern the same methods break down at far smaller shares of missing data. At the heaviest level none of them brings the relative covariance error below 0.6, the first principal component falls from 41.74% to about 25%, and the portfolios end at little more than half of the oracle’s cumulative return, with the five imputers indistinguishable from one another. What has been deleted there is not a random part of the sample but the part that carries the covariance structure, and none of the methods tested could reconstruct from the observed panel what the observed panel no longer contains. Two qualifications belong with that reading. The stressed pattern changes the selection rule and the geometry of the missingness at the same time, deleting preferentially on turbulent dates and deleting date by date across assets, so the present design does not separate the contribution of the two; and the comparison rests on a single panel and one run per configuration. Within those limits, a comparable share of missing cells was not a comparable problem.

The deep models did not earn their cost here. EM is rarely beaten in the metric tables and never by a wide margin, CSDI matches it, and BRITS falls behind at two of the stressed levels. EM fills a panel in about a second, while CSDI needs some twenty-four minutes of sampling after four hours of training, four orders of magnitude more for a reconstruction that is not better. The comparison is tied to the training budget and the

hardware of Section 3.3, but the size of the gap leaves a good deal of room before the conclusion would turn.

Pointwise accuracy is a poor proxy for the quality of an imputed panel. Under the MCAR patterns the MAE of zero imputation is only about fifty percent worse than the best, while its covariance error is close to an order of magnitude worse than EM's, so a method can look acceptable entry by entry and still distort the geometry of the panel. The distortion is also systematic rather than noisy. Under the tested MCAR configurations every conditional imputer raises the share of variance carried by the first factor, from 41.74% to between 45.6% and 48.5% at sparse rate 0.3, because the value it inserts is a conditional location estimate and therefore lies closer to the dominant directions of the panel than a real return does. The more entries are filled, the more the panel comes to look like its own factor model, and no pointwise metric registers this. This is a property of filling with point estimates rather than with draws, so it should be expected to persist for point imputation generally, but the evidence here is for these methods under these patterns.

The rankings largely survive downstream, and the exceptions carry the most information. Low-rank completion is among the most accurate methods entry by entry and has the largest five-factor subspace distance at ten of the fifteen levels, reaching 0.39 at sparse rate 0.3 against 0.21 for EM, since truncating the spectrum discards exactly those weakly separated directions inside the leading subspace. Complete-case deletion reads as merely inefficient in the metric tables and produces allocations whose Sharpe ratio falls to almost zero at sparse rate 0.1 and whose mean return turns negative at the gap level (1000, 20). The two downstream evaluations are themselves connected: since $\hat{\Sigma}^{-1} = \sum_i \hat{\lambda}_i^{-1} \hat{v}_i \hat{v}_i^\top$, the portfolio weights each direction by the reciprocal of its variance and so loads most heavily on the trailing eigenvectors, which are the directions the imputations recover worst. The error maximisation seen in Section 3.1 is that distortion viewed through an inverse that magnifies it.

The practical reading is that an imputation should be scored against the use it is put to. Reconstruction error orders the methods correctly at the easy levels, where the ordering hardly matters, and is least informative exactly where the choice of method is consequential.

4.2 Limitations

The evidence comes from a single panel: fifty liquid large-cap equities from one market over one period, at daily frequency. Fifty assets against 4125 dates is a comfortable ratio for covariance estimation, and a cross section closer in size to the sample length would make every estimate less stable and would likely widen the differences reported here. The missingness is injected rather than observed, so its mechanism is known and

stationary, whereas real gaps arise from halts, delistings and illiquidity and follow no law the experimenter can write down; the stressed pattern is a stylised form of MNAR and not a model of any particular cause; and because it deletes preferentially on turbulent dates and does so date by date across assets, the selection rule and the date-aligned geometry it induces are not separated by this design. Each configuration was run once, so the tables carry no estimate of sampling variation around the numbers they report. The EM results are those of the Gaussian model, so the comparison with the neural imputers is a comparison with a classical estimator that was not given the robust Student- t treatment its heavy-tailed data would justify. The two downstream evaluations both read the covariance matrix, which makes them complementary but not independent tests, and the portfolio evaluation is in sample, a diagnostic of reconstruction fidelity rather than a statement about investment performance. Finally, the experiments are run in an offline reconstruction setting in which every method may use full-sample information. BRITS reads the panel in both directions and CSDI conditions on every observed value, so the results describe reconstruction after the fact and not what would be available to a user filling a gap in real time, with only the past in hand.

4.3 Future work

The last of those limitations is the natural continuation. Restricting each imputer to the information available up to the date being filled turns the problem from smoothing into a filtering one and makes the cost of that restriction measurable, which is what a user working in real time needs to know and what the benchmarks in the literature do not report. A second direction is to widen the downstream evaluations beyond the second moment, with a Value-at-Risk backtest, a hedging exercise or a trading rule, since a criterion that does not pass through the covariance would test parts of the reconstruction that neither of the present evaluations reaches. A third is to carry the uncertainty of the imputation forward instead of discarding it. CSDI produces a predictive distribution that we collapsed to a median, and the inflation of the factor structure documented above is the artefact that filling with point estimates would be expected to create, so a multiple-imputation treatment in which the downstream decision sees the spread rather than a point estimate is the natural test of that explanation. Beyond these, the same design applies directly to larger cross sections, lower frequencies and other asset classes, and eventually to panels whose gaps are real rather than injected, where the mechanism has to be inferred rather than assumed.

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